



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
 - Data Collection:
 - Used SpaceX API and web scraping.
 - Data Wrangling:
 - Cleaned and transformed data.
 - EDA:
 - Explored data with visualizations and SQL.
 - Interactive Analytics:
 - Created Folium maps and Plotly Dash dashboard.
 - Predictive Modeling:
 - Built classification models for launch outcomes.
- Summary of all results
 - Data Insights:
 - Analyzed 101 launches.
 - KSC LC-39A had the highest success rate.
 - Payload mass influenced success.
 - Interactive Dashboard:
 - Showcased launch success and payload vs. outcome plots.
 - Predictive Analysis:
 - Developed accurate classification model.
 - Conclusion:
 - Identified key factors for success.
 - Provided insights for future launches.

Introduction

- Project background and context
 - Analyzing SpaceX launch data to understand success and failure factors.
 - SpaceX's extensive launch data provides a rich dataset for analysis.
 - This project aims to uncover insights to optimize future SpaceX launches and improve mission success rates.
- Key Questions:
 - Success Factors:
 - What determines the success or failure of a launch?
 - Site Performance:
 - How do different launch sites compare in success rates?
 - Payload Impact:
 - How does payload mass affect launch outcomes?
 - Predictive Modeling:
 - Can we predict the success of future launches?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Utilized SpaceX API and web scraping to gather detailed launch data.
- Perform data wrangling
 - Cleaned and processed data to handle missing values and ensure consistency.
- Perform exploratory data analysis (EDA) using visualization and SQL
 - Conducted EDA using visualizations and SQL queries to uncover patterns and insights.
- Perform interactive visual analytics using Folium and Plotly Dash
 - Created interactive maps with Folium to visualize launch site locations and outcomes.
 - Developed dynamic dashboards with Plotly Dash for deeper data exploration.
- Perform predictive analysis using classification models
 - Built classification models to predict launch outcomes.
 - Tuned and evaluated models to identify the best-performing one.

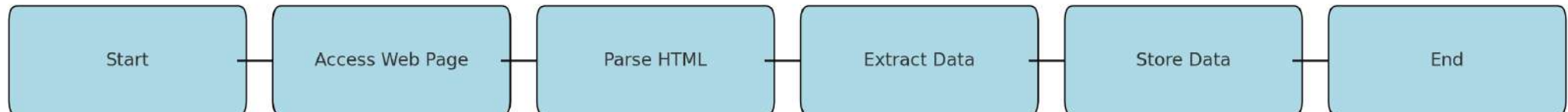
Data Collection

- SpaceX API:
 - Collected detailed launch data including launch dates, sites, outcomes, and payload information.
 - Used RESTful API calls to retrieve data programmatically.
- Web Scraping:
 - Supplemented API data with additional details from SpaceX website and other sources.
 - Employed BeautifulSoup and Selenium for automated data extraction.

API Data Collection

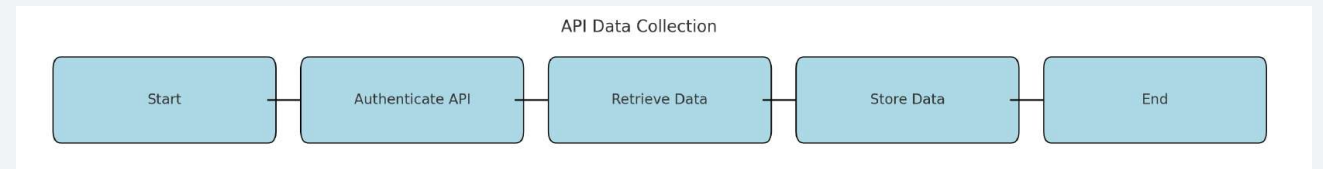


Web Scraping



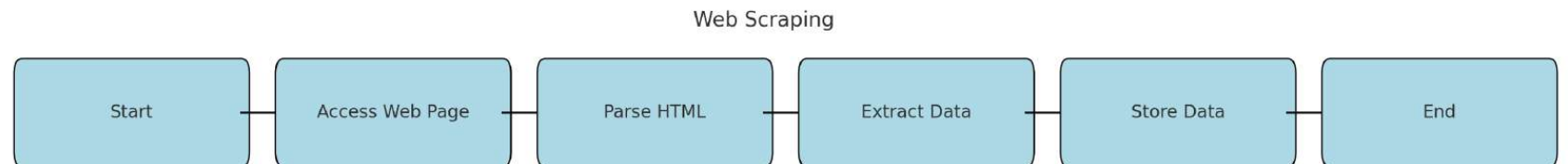
Data Collection – SpaceX API

- Collected detailed launch data including launch dates, sites, outcomes, and payload information.
- Used RESTful API calls to retrieve data programmatically.
- <https://github.com/mabrurh/IBMDDataScienceProfessionalCertificateCapstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>



Data Collection - Scraping

- Supplemented API data with additional details from SpaceX website and other sources.
- Employed BeautifulSoup and Selenium for automated data extraction.
- <https://github.com/mabrurh/IBMDDataScienceProfessionalCertificateCapstone/blob/main/jupyter-labs-webscraping.ipynb>



Data Wrangling

- 1 Load Data:
 - Read data from CSV, JSON, and API sources.
- 2 Clean Data:
 - Handle missing values and remove duplicates.
- 3 Transform Data:
 - Convert data types and normalize fields.
- 4 Integrate Data:
 - Merge data from different sources for a comprehensive dataset.
- 5 Store Processed Data:
 - Save cleaned and transformed data for analysis.

<https://github.com/mabrurh/IBMDataScienceProfessionalCertificateCapstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

Data Wrangling Process



EDA with SQL

SQL Queries Performed:

- 1) Identified all distinct launch sites.
- 2) Retrieved records where launch site names begin with 'CCA'.
- 3) Calculated the total payload mass carried by NASA boosters.
- 4) Computed the average payload mass for boosters of version F9 v1.1.
- 5) Found the date of the first successful landing on a ground pad.
- 6) Listed boosters that landed on a drone ship with payload mass between 4000 and 6000 kg.
- 7) Calculated the total number of successful and failed mission outcomes.
- 8) Identified boosters that carried the maximum payload mass.
- 9) Listed failed landing outcomes on drone ships for 2015, including booster versions and launch site names.
- 10) Ranked landing outcomes between specific dates in descending order.

<https://github.com/mabrurh/IBMDDataScienceProfessionalCertificateCapstone/blob/main/edadataviz.ipynb>

Build an Interactive Map with Folium

Map Objects Created:

- Markers:
 - Placed markers at each launch site location.
 - Purpose: To visually identify and distinguish the geographical positions of different launch sites.
- Circles:
 - Added colored circles to represent the outcomes of launches (green for success, red for failure).
 - Purpose: To provide a quick visual reference for the success rate and performance of each launch site.
- Polylines:
 - Used lines to connect launch sites to their nearest major landmarks (e.g., highways, railways).
 - Purpose: To analyze the accessibility and logistical considerations of each launch site.
- Popups:
 - Integrated popups with detailed information about each launch site and its respective launches.
 - Purpose: To give users immediate access to essential data points without leaving the map.

https://github.com/mabrurh/IBMDDataScienceProfessionalCertificateCapstone/blob/main/lab_jupyter_launch_site_location.ipynb 12

Build a Dashboard with Plotly Dash

Plots/Graphs Added:

- Launch Success Count Pie Chart:
 - Shows overall success and failure rates.
 - Purpose: Provides a clear overview of launch success rates.
- Launch Success by Site Pie Chart:
 - Displays success rates for each launch site.
 - Purpose: Highlights individual site performance.
- Payload vs. Launch Outcome Scatter Plot:
 - Interactive plot with range slider for payload mass.
 - Purpose: Analyzes impact of payload mass on launch success.
- Interactive Controls:
 - Dropdowns and sliders for site selection and payload filtering.
 - Purpose: Enables dynamic data exploration and customization.

https://github.com/mabrurh/IBMDataScienceProfessionalCertificateCapstone/blob/main/spacex_dash_app.py 13

Predictive Analysis (Classification)

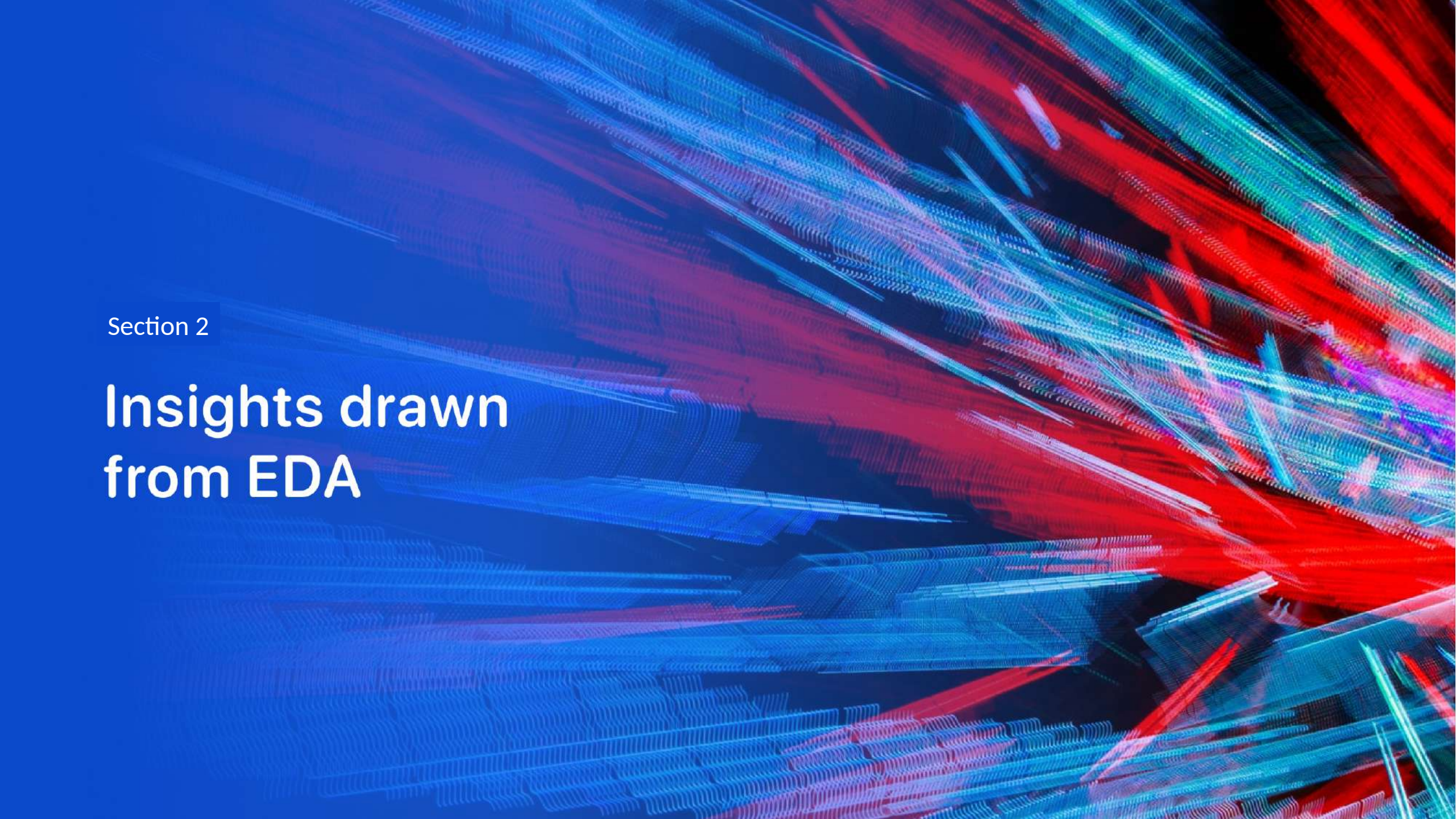
Model Development Process:

- 1 Data Preparation:
 - Cleaned and preprocessed data for modeling.
- 2 Feature Selection:
 - Identified relevant features impacting launch success.
- 3 Model Building:
 - Used algorithms like Logistic Regression, Decision Trees, and Random Forests.
- 4 Model Evaluation:
 - Assessed models using accuracy, precision, recall, and F1 score.
- 5 Model Improvement:
 - Tuned hyperparameters using GridSearchCV.
 - Applied cross-validation for robustness.
- 6 Best Model Selection:
 - Selected the model with the highest accuracy and balanced metrics.

https://github.com/mabrurh/IBMDDataScienceProfessionalCertificateCapstone/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

- Exploratory data analysis results
 - Identified KSC LC-39A as the launch site with the highest success rate.
 - Discovered that payload mass significantly impacts launch success.
- Interactive analytics demo in screenshots
- Predictive analysis results
 - After testing with Logistic Regression, Support vector machines (SVM), Decision trees and K Nearest Neighbors (KNN), It was found that Logistic regression was the best estimator of landing success with an accuracy of 83.34%
 - False Positives seemed to be the biggest problem when predicting.

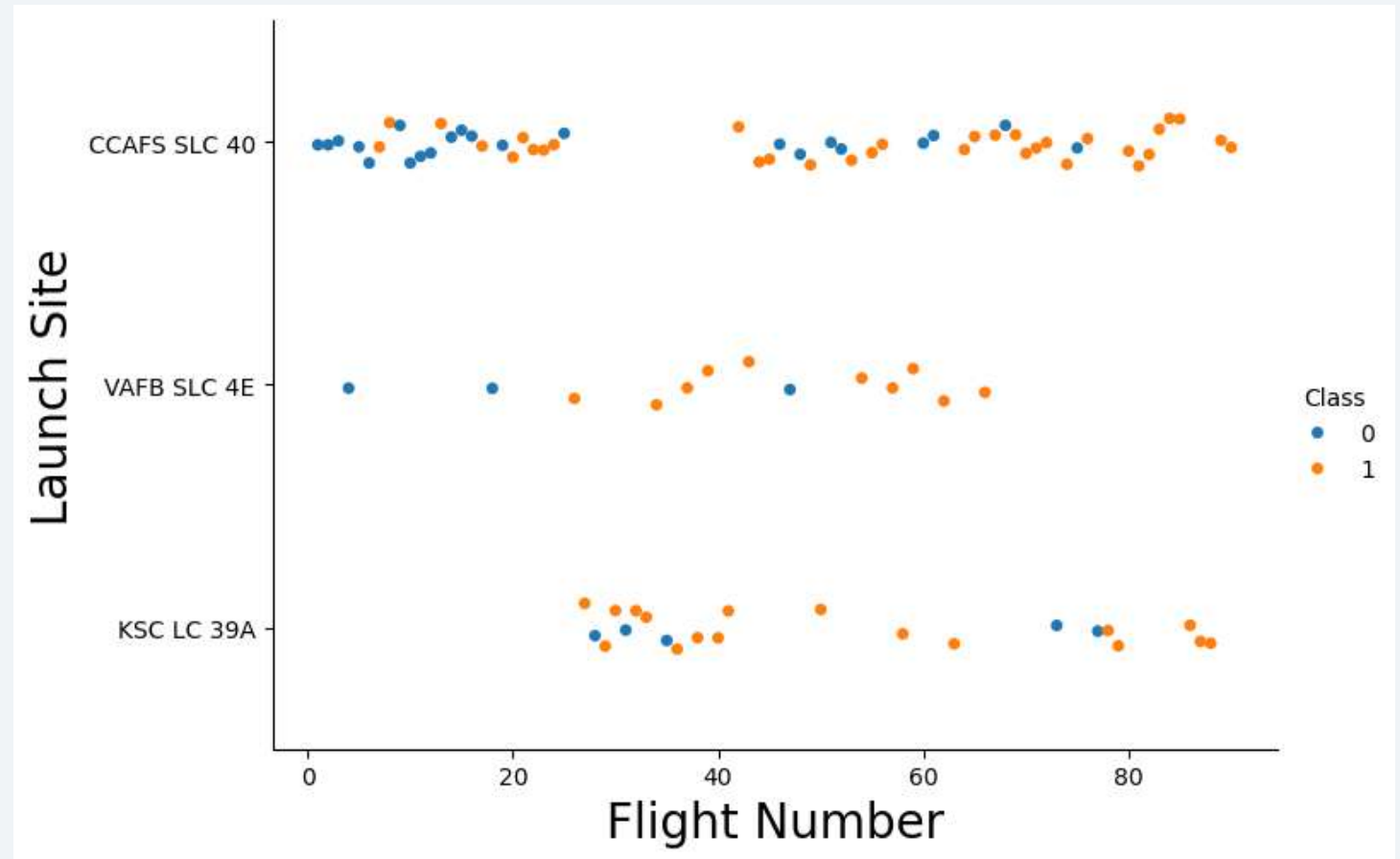
The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a dynamic pattern of diagonal streaks in shades of blue and red on the right. Overlaid on these streaks is a faint, light-blue grid pattern, giving the impression of a digital or data-driven environment.

Section 2

Insights drawn from EDA

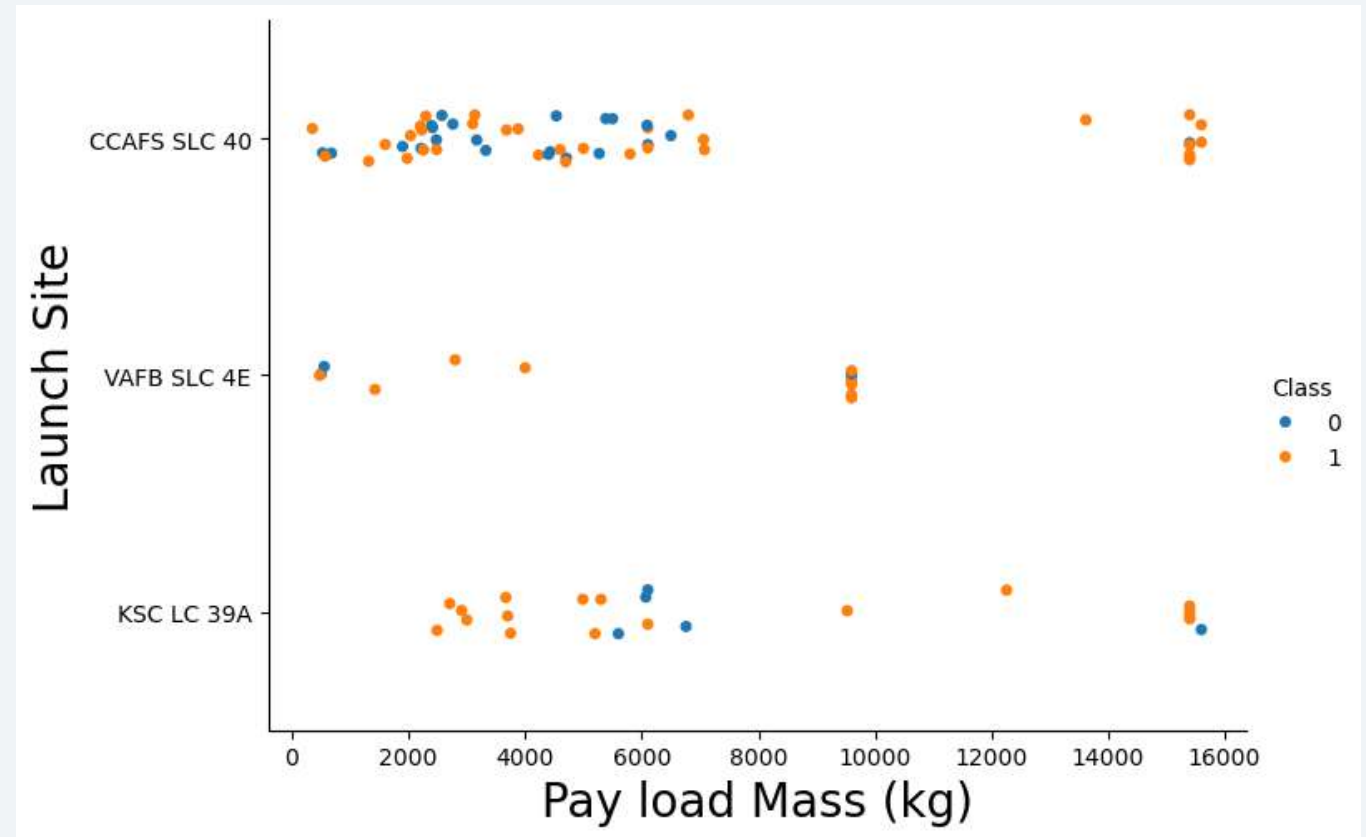
Flight Number vs. Launch Site

The scatter plot shows that CCAFS SLC 40 had the most number of flights and the latest flight numbers for each show a much higher success rate for each launch site (P1).



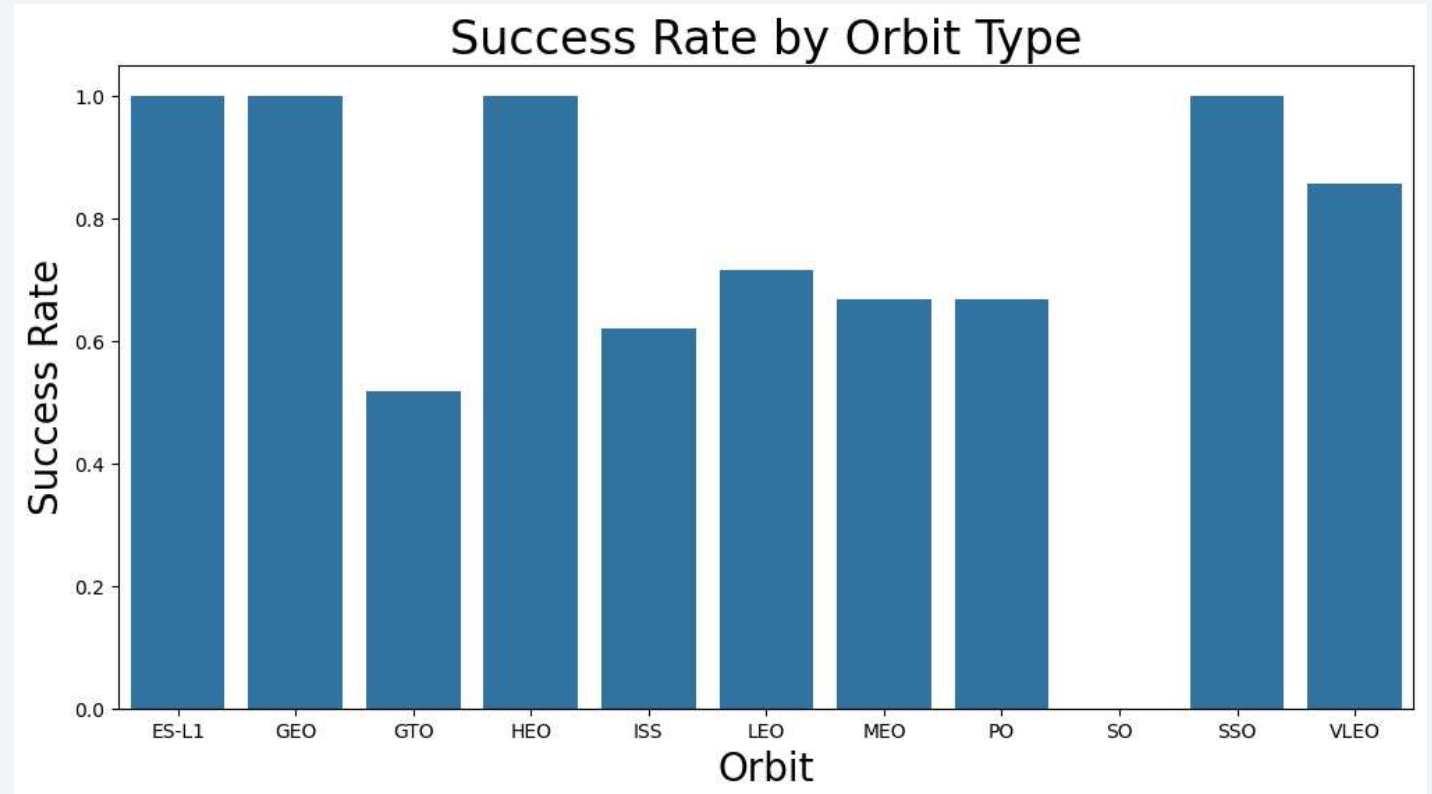
Payload vs. Launch Site

The scatter plot shows that most launches had a payload of less than 7000kg, and a higher success rate for higher payloads as perhaps more was at risk (P2).



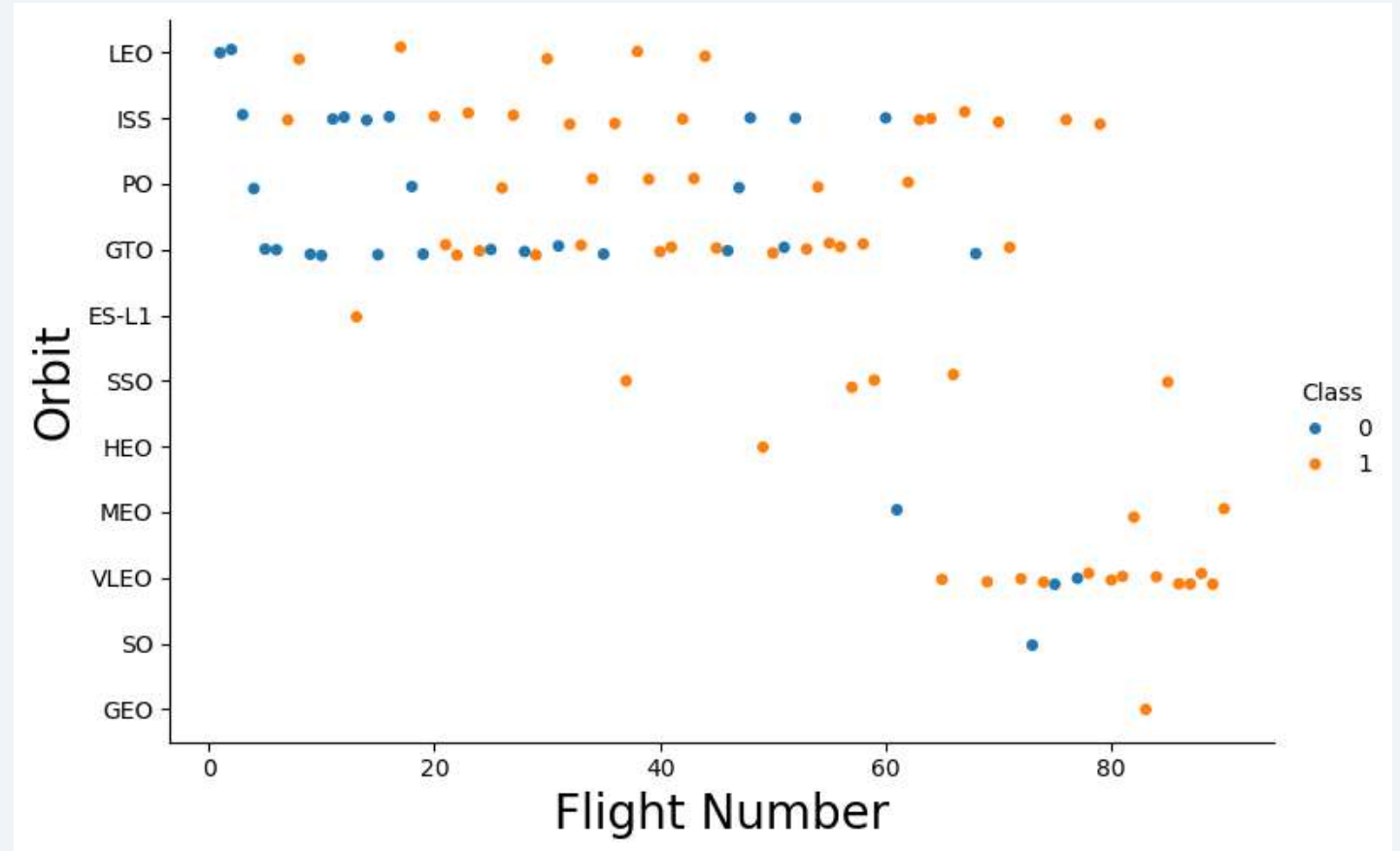
Success Rate vs. Orbit Type

The bar chart shows that the least successful Orbit type was GTO and the most successful with no failures were SSO, tied with HEO, GEO and ES-L1 (P3).



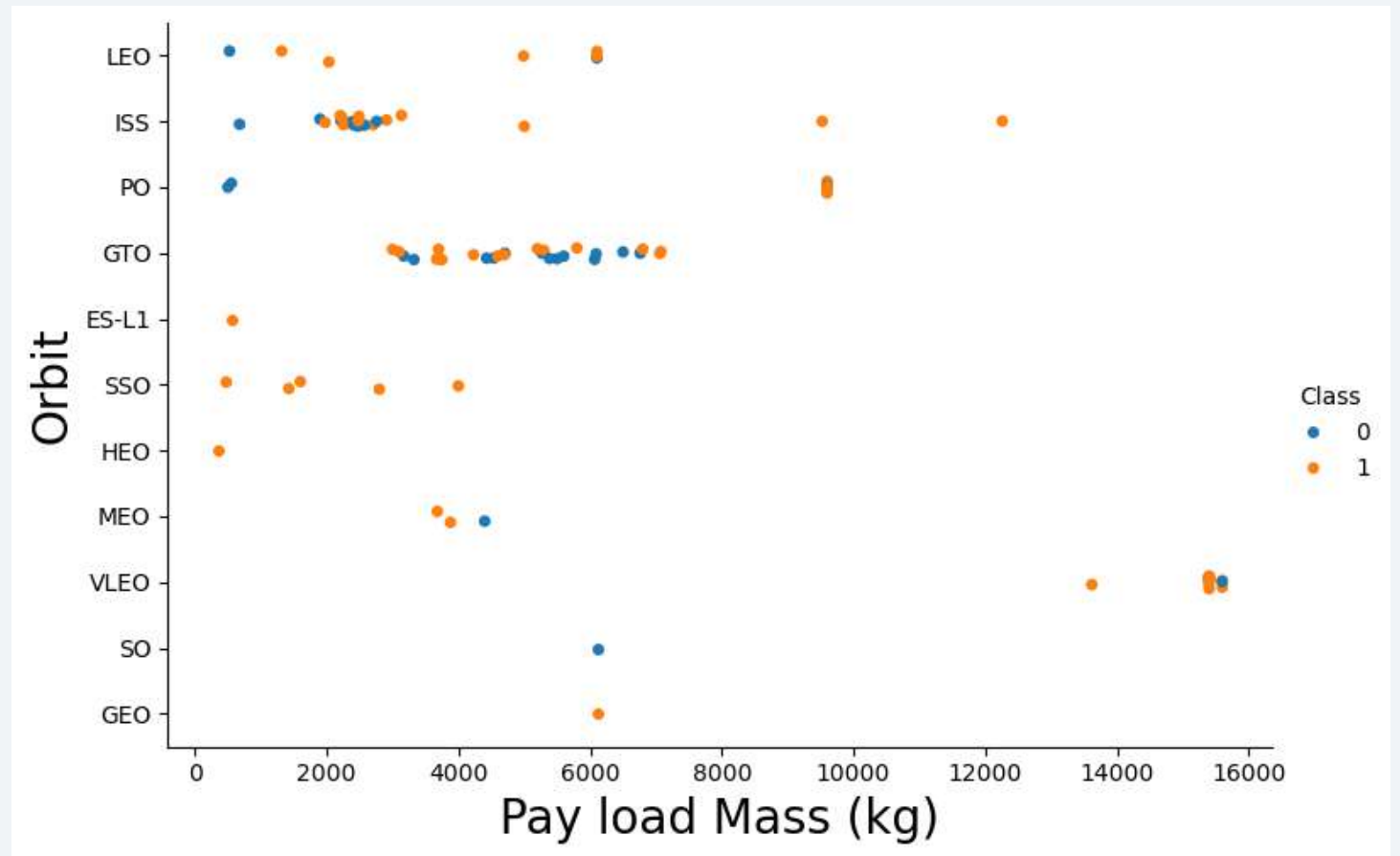
Flight Number vs. Orbit Type

The scatter plot shows that the orbit type with most successful flights was SSO and the least successful was GTO (P4).



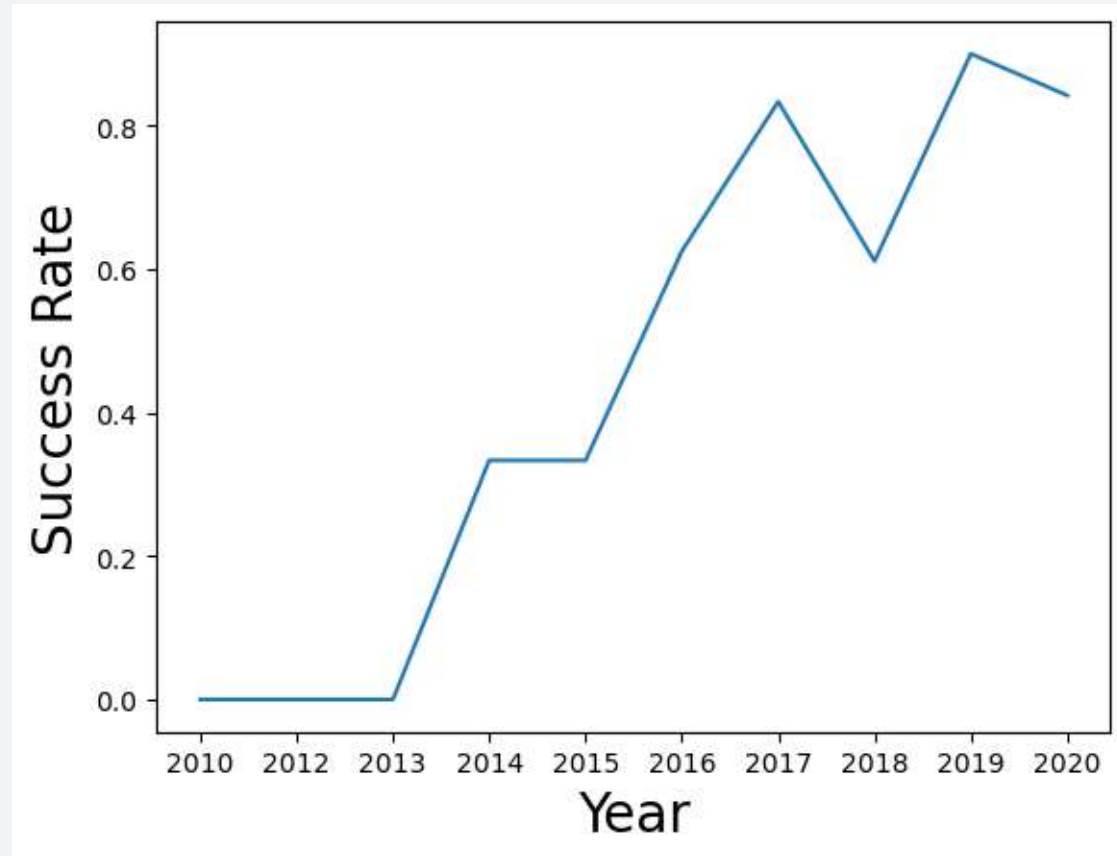
Payload vs. Orbit Type

The scatter plot shows that higher payloads are generally accompanied by successful flights with the most successful orbit type being SSO once again, however it was only used for smaller payloads (4000kg or less) (P5).



Launch Success Yearly Trend

As expected and as revealed by the previous flight no plots, the success rate of flights has increased year over year. There was a drop in 2017, but SpaceX must have learned from it's mistakes as launch success increased again the year after (P6).



All Launch Site Names

The launch sites are:

- CCAFS LC-40
- VAFB SLC-4E
- KSC LC-39A
- CCAFS SLC-40

Used SELECT DISTINCT Statement to find all distinct Launch site names from the Launch_Site column in the dataset (S1).

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Used the LIKE statement with CCA% to find launch sites that begin with CCA (S2).

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

The total payload mass was 48213 Kg.

Used the SUM statement to calculate the sum of all payload masses from the Payload mass column (S3).

TotalPayloadMass
48213

Average Payload Mass by F9 v1.1

The average Payload mass was 2928.4 Kg.

Used the AVG statement to calculate the average and the where statement to indicate which records' payload masses in particular to average (S4).

AveragePayloadMass
2928.4

First Successful Ground Landing Date

The first successful ground landing date was 22 December, 2015.

Used the MIN statement to find the first date and WHERE statement to search for Successful ground landing (S5).

FirstSuccessfulLanding

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

The booster versions were:

- F9 FT B1022
- F9 FT B1026
- F9 FT B1021.2
- F9 FT B1031.2

Used the WHERE statement to find the names of boosters which have successfully landed on drone ship and the BETWEEN statement to find those that had payload mass greater than 4000 but less than 6000 (S6).

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

It seems out of 101 missions, 100 were successful and 1 was a failure.

Used the COUNT statement to calculate the total number of successful and failure mission outcomes then grouped by outcome (S7).

Mission_Outcome	OutcomeCount
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

Used the SELECT, WHERE and MAX statements to find the Boosters that carried the maximum payloads (S8).

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

Used the AND statement to list the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015 (S9).

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Used the BETWEEN and GROUP BY ... DESC statements to rank the count of landing outcomes between the dates 2010-06-04 and 2017-03-20, in descending order (S10).

Landing_Outcome	OutcomeCount
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a deep blue, with the horizon line visible. The city lights are concentrated in the lower right quadrant, showing a dense network of urban areas. The text "Section 3" is overlaid on the left side of the image.

Section 3

Launch Sites Proximities Analysis

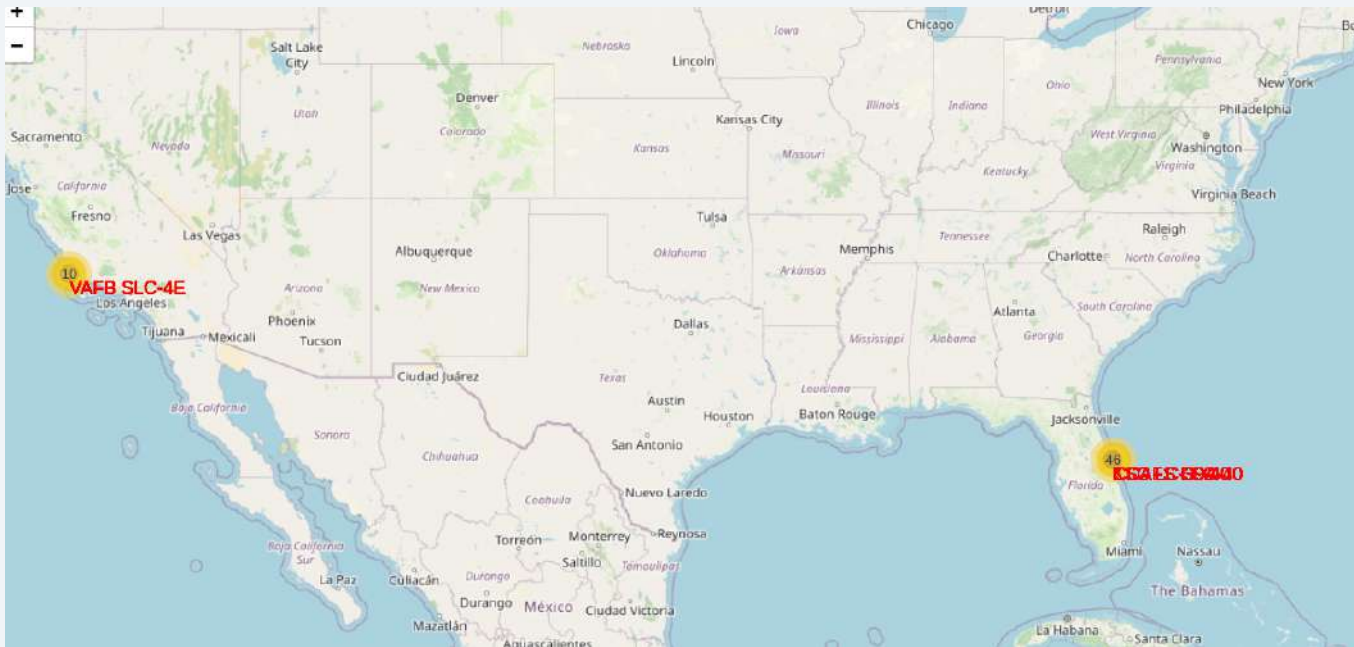
All Launch sites

All launch sites are located in coastal regions of the USA. With a few on the east coast and one on the west coast.



Launch Outcome Map

Most launches originate from the east coast. Most launch sites have a majority of failed launches except for one, KSC LC-39A which is the Kennedy Space Center.



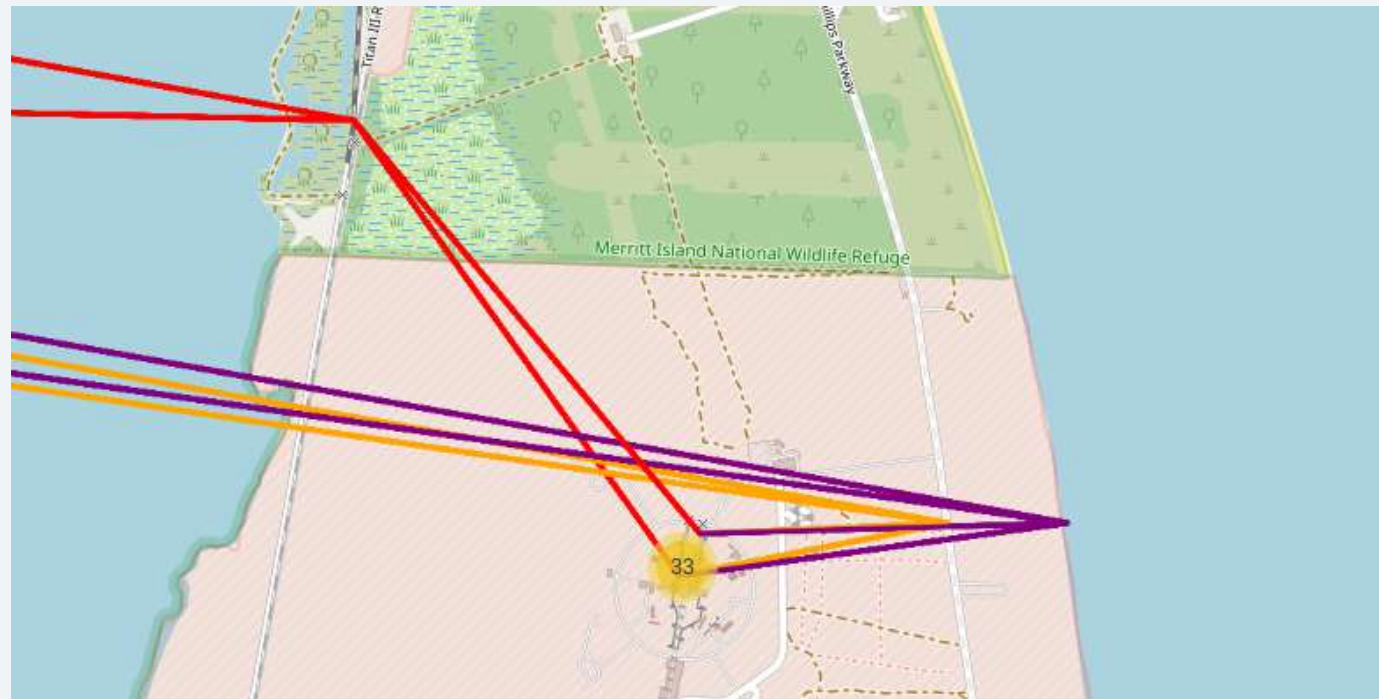
Proximity to railways, highways and coastlines

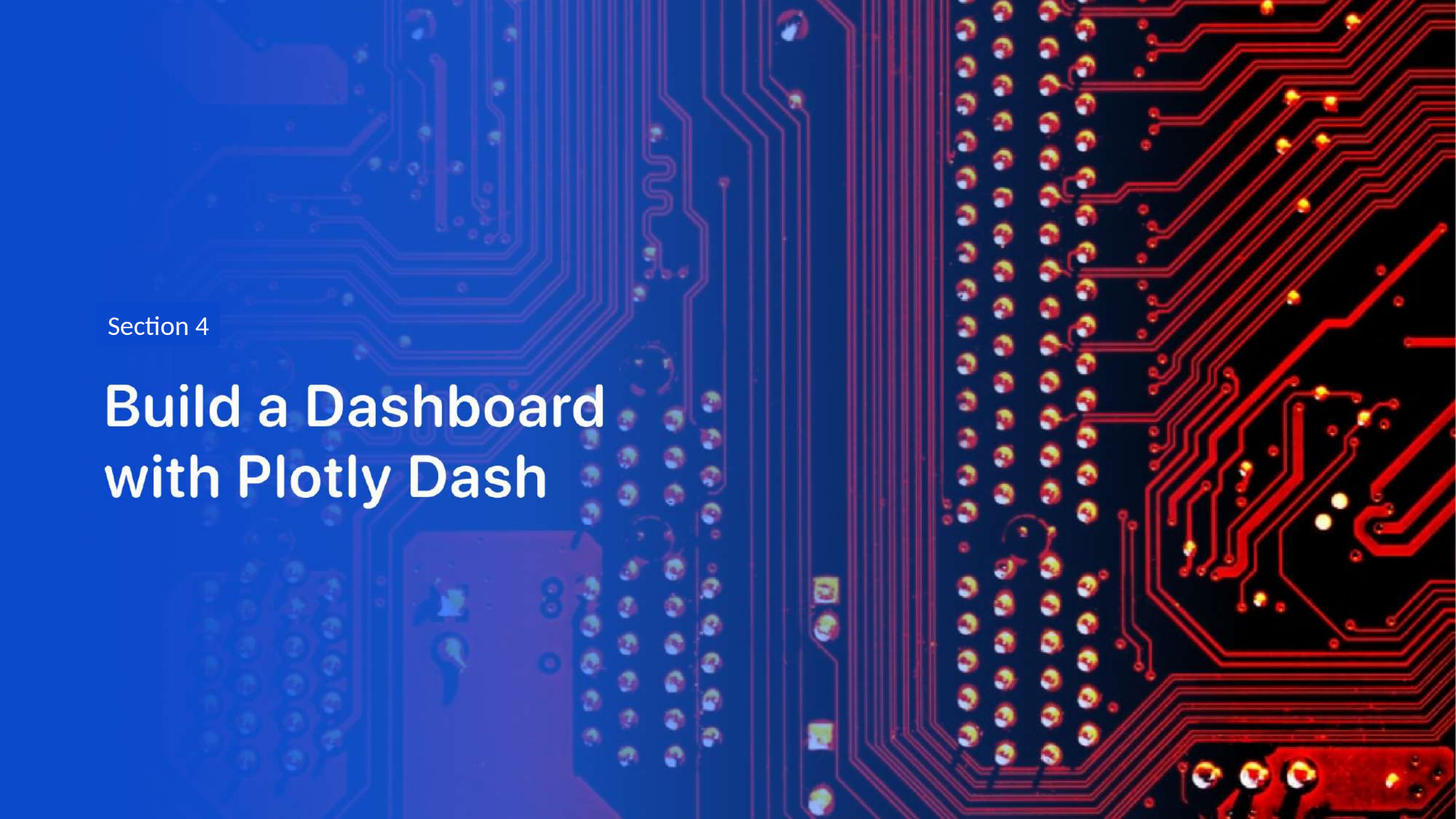
The launch sites are generally in close proximity to railways, as indicated by the red lines. This is likely for ease of transportation of heavy and large rocket components.

The launch sites are also close to highways (orange lines), ensuring efficient ground transportation routes for both materials and personnel.

The proximity to coastlines (purple lines) is evident, which is essential for safety reasons, allowing rockets to be launched over open water to minimize risk to populated areas.

The launch sites are typically located away from major cities, reducing the risk and impact of potential accidents or failures on densely populated areas. These proximities highlight the strategic planning involved in choosing launch site locations.





Section 4

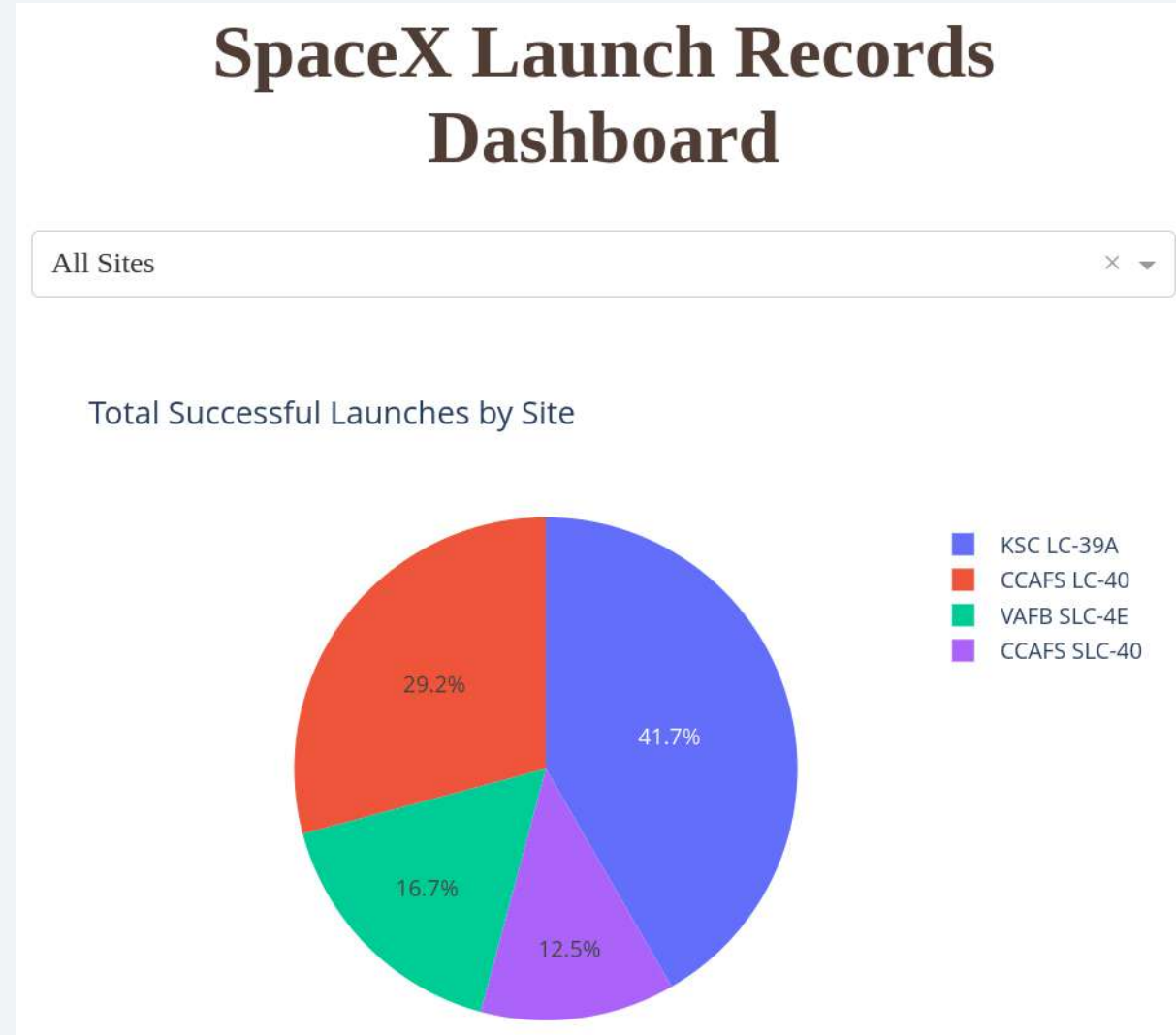
Build a Dashboard with Plotly Dash

Pie chart of Launch Success count for all sites

The pie chart shows that most successful launches were from the launch site KSC LC-39A.

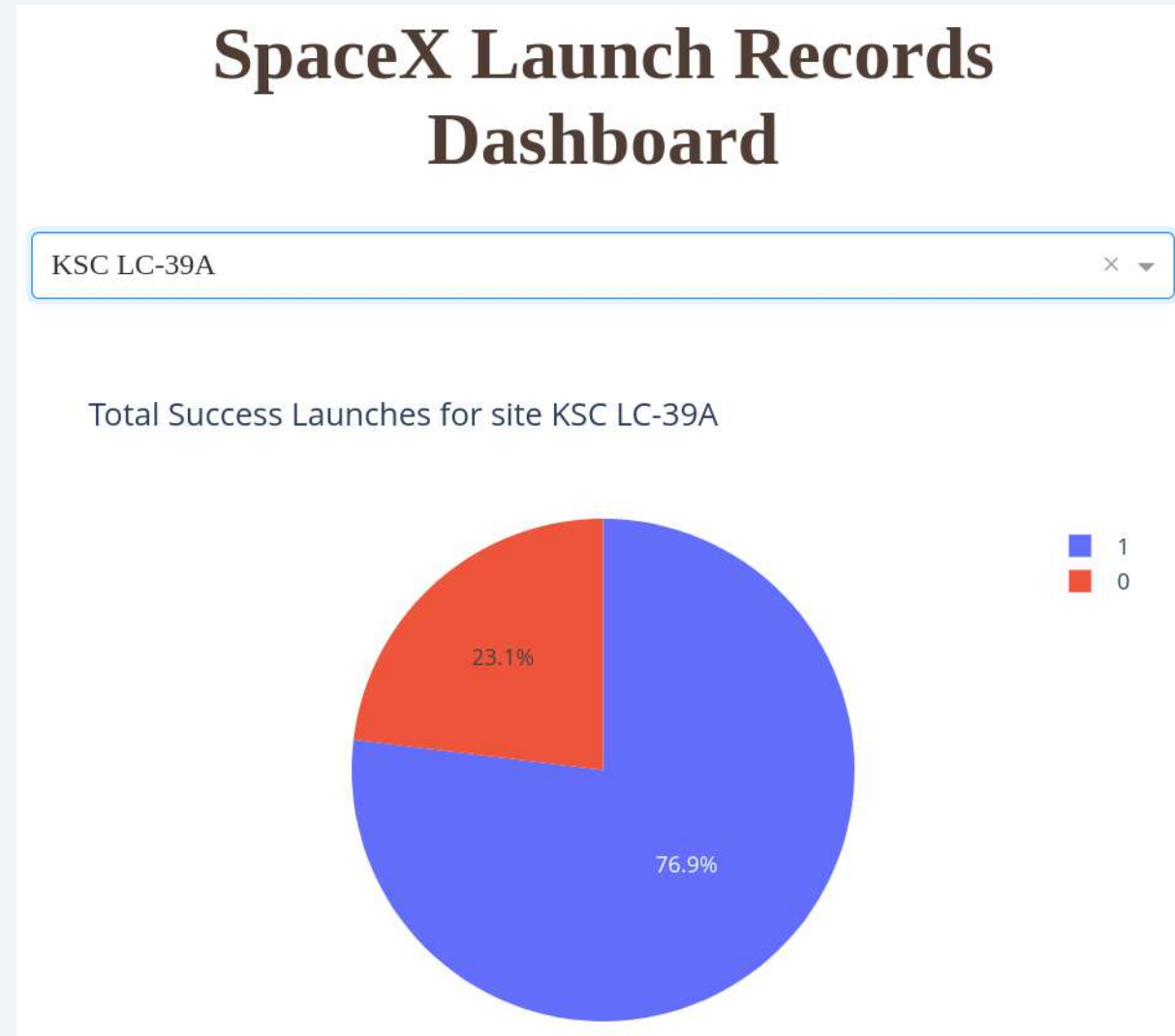
Ranked in descending order of successful launches, the launch sites are ranked:

- 1) KSC LC-39A
- 2) CCAFS LC-40
- 3) VAFB SLC-4E
- 4) CCAFS SLC-40



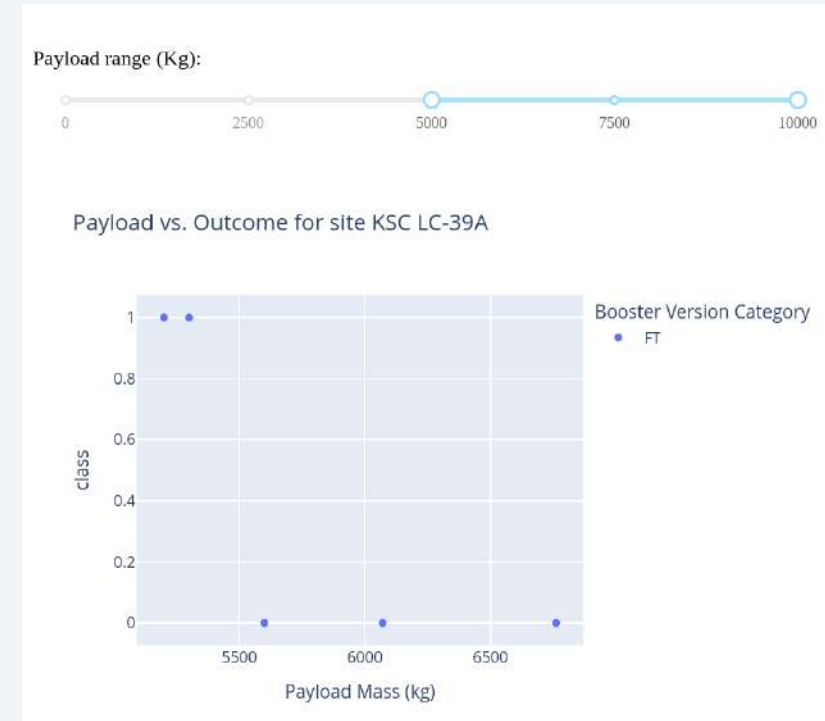
Pie Chart for Launch site with highest success rate

The launch site KSC LC-39A had the highest success rate among all launch sites in the dataset. 76.9% of the launches from KSC LC-39A were successful and 23.1% were failures.



Payload vs launch outcome Scatter plot

The scatter plots show that for lower payload ranges (5000Kg and below), All launches were successful. On the other hand, for higher payload ranges, (Between 5000Kg and 10000Kg), there is a much lower success rate as payload mass increases.



Section 5

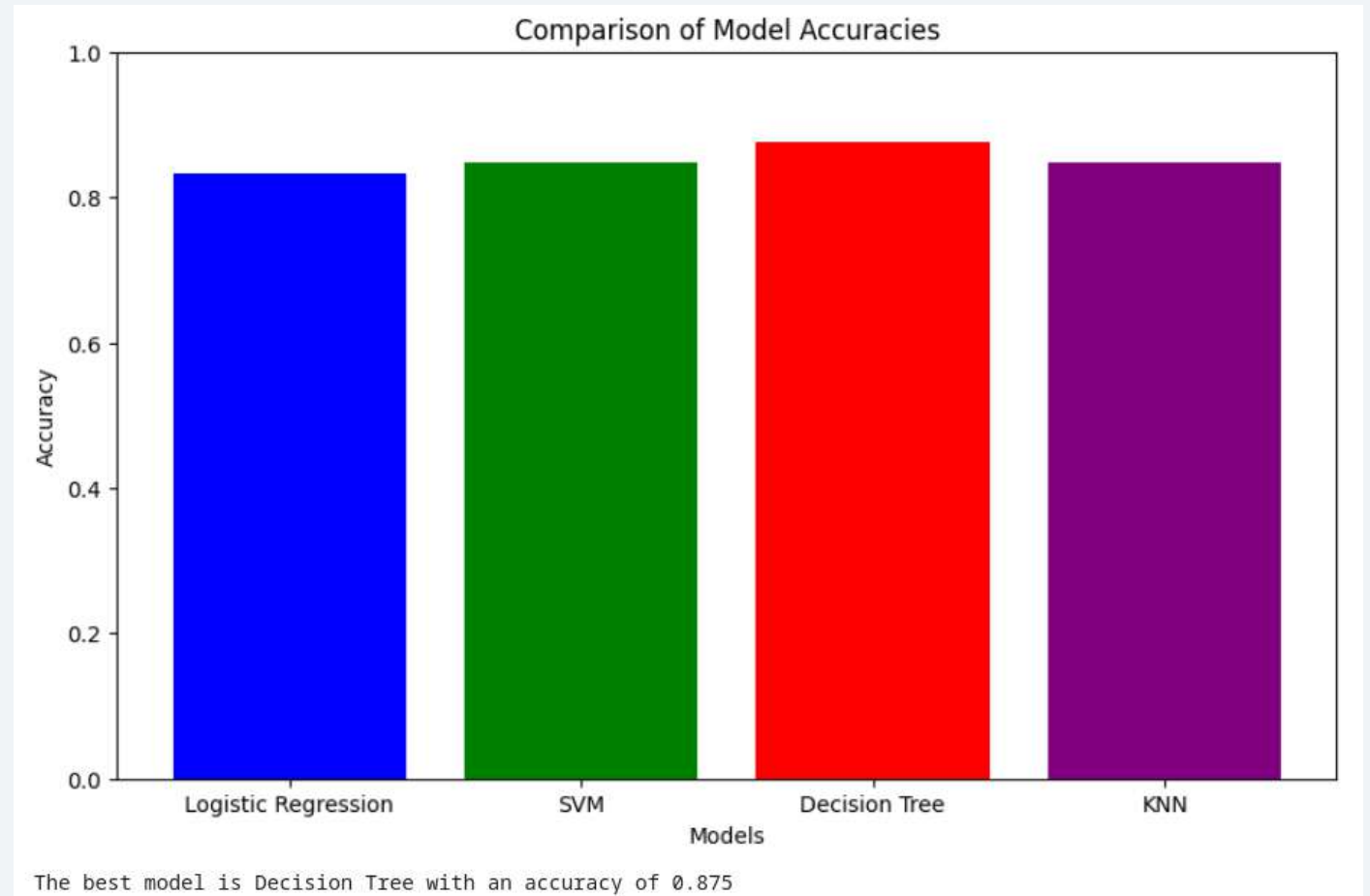
Predictive Analysis (Classification)

Classification Accuracy

The best model was Decision Tree with an accuracy of 0.875

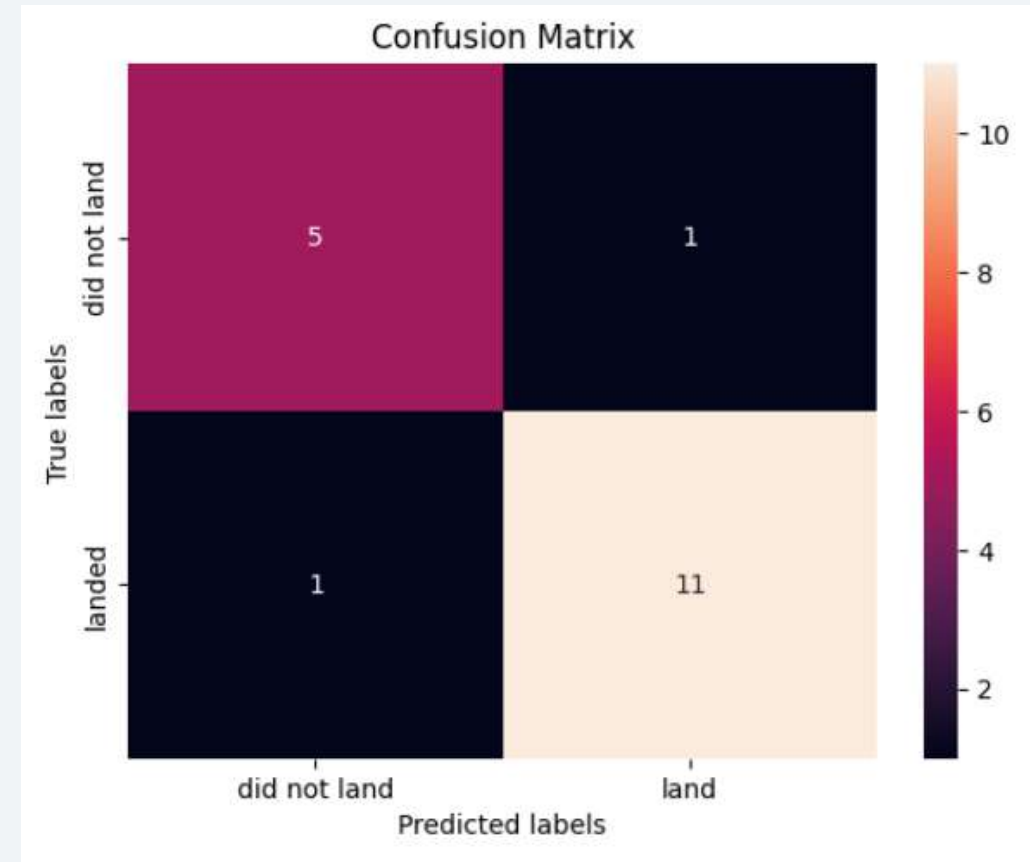
The accuracies of all the models in descending order are the following:

- 1) Decision Tree test accuracy: 0.875
- 2) KNN test accuracy: 0.8482142857142858
- 3) SVM test accuracy: 0.8482142857142856
- 4) Logistic Regression test accuracy: 0.8333333333333334



Confusion Matrix

- The best performing model was found to be Decision Trees.
- The confusion matrix shows that the major issue was false positives and false negatives, indicated by the number 1 in the top right and bottom left quartiles showing intersections between “land” and “did not land”



Conclusions

- Key Success Factors:
 - Launch site and payload mass significantly influence launch success.
 - KSC LC-39A has the highest success rate among all sites.
- EDA Insights:
 - The most successful orbit type is SSO and the least successful is GTO.
 - Higher payloads generally correspond to higher success rates.
- Interactive Analytics:
 - Folium maps and Plotly Dash dashboard provided valuable interactive tools.
 - Enabled detailed exploration and filtering of launch data.
- Predictive Modeling:
 - Developed several classification models for predicting launch outcomes.
 - Decision Tree model performed best with an accuracy of 87.5%.
- Data-Driven Decisions:
 - Insights from this analysis can guide future SpaceX launch strategies.
 - Data-driven approaches can optimize success rates and operational efficiency.

Appendix – Python

- 1) `sns.catplot(y="LaunchSite", x="FlightNumber", hue="Class", data=df, aspect = 5)`
`plt.xlabel("Flight Number",fontSize=20)`
`plt.ylabel("Launch Site",fontSize=20)`
`plt.show()`
- 2) `sns.catplot(y="LaunchSite", x="PayloadMass", hue="Class", data=df, aspect = 5)`
`plt.xlabel("Pay load Mass (kg)",fontSize=20)`
`plt.ylabel("Launch Site",fontSize=20)`
`plt.show()`
`%sql SELECT SUM(PAYLOAD__MASS__KG_) as TotalPayloadMass FROM SPACEXTABLE WHERE Customer LIKE 'NASA (CRS)%'UU`
- 3) `orbit_success_rate = df.groupby('Orbit')['Class'].mean().reset_index()`
`plt.figure(figsize=(12, 6))`
`sns.barplot(x='Orbit', y='Class', data=orbit_success_rate)`
`plt.xlabel('Orbit', fontsize=20)`
`plt.ylabel('Success Rate', fontsize=20)`
`plt.title('Success Rate by Orbit Type', fontsize=24)`
`plt.show()`
`%sql SELECT MIN(Date) as FirstSuccessfulLanding FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)'`

Appendix – Python

- 4) `sns.catplot(y="Orbit", x="FlightNumber", hue="Class", data=df, aspect = 5)`
`plt.xlabel("Flight Number",fontsize=20)`
`plt.ylabel("Orbit",fontsize=20)`
`plt.show()`
- 5) `sns.catplot(y="Orbit", x="PayloadMass", hue="Class", data=df, aspect = 5)`
`plt.xlabel("Pay load Mass (kg)",fontsize=20)`
`plt.ylabel("Orbit",fontsize=20)`
`plt.show()`
- 6) `year_success_rate = df.groupby('Date')['Class'].mean().reset_index()`
`sns.lineplot(y="Class", x="Date", data=year_success_rate)`
`plt.xlabel("Year",fontsize=20)`
`plt.ylabel("Success Rate",fontsize=20)`
`plt.show()`

Appendix – SQL

- 1) %sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE
- 2) %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
- 3) %sql SELECT SUM(PAYLOAD_MASS__KG_) as TotalPayloadMass FROM SPACEXTABLE WHERE Customer LIKE 'NASA (CRS)%'UU
- 4) %sql SELECT AVG(PAYLOAD_MASS__KG_) as AveragePayloadMass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1'
- 5) %sql SELECT MIN(Date) as FirstSuccessfulLanding FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)'
- 6) %sql SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 5999
- 7) %sql SELECT Mission_Outcome, COUNT(*) as OutcomeCount FROM SPACEXTABLE GROUP BY Mission_Outcome
- 8) %sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)
- 9) %sql SELECT substr(Date, 6, 2) as Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE Landing_Outcome = 'Failure (drone ship)' AND substr(Date, 1, 4) = '2015'
- 10) %sql SELECT Landing_Outcome, COUNT(*) as OutcomeCount FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY OutcomeCount DESC

Thank you!

